gpeyt1-12 · Sand-cone relative compaction

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

A field density test on a compacted embankment lift: the mass of calibrated sand that filled the hole (after subtracting the cone correction) was 3.36 lb; the bulk density of the calibrated sand is 93.5 pcf. The moist soil removed from the hole weighed 4.66 lb, and its oven-dry water content is 11.8%. The lab standard Proctor gave $\gamma_{d,max}$ = 121.0 pcf at w_{opt} = 10.5%. Specification: ≥ 95% relative compaction, w within −1/+3% of optimum. Lift thickness 8 in.

Most nearly, the relative compaction and the test result are:

A. 107.2%, passes
B. 94.5%, fails
C. 95.9%, passes
D. 104.3%, passes

Given equations / data

> $V = W_{sand}/\gamma_{sand}$; $\gamma_t = W/V$; $\gamma_d = \gamma_t/(1 + w)$; $RC = \gamma_d/\gamma_{d,max} \times 100\%$

Worked solution

1. Hole volume $V = 3.36/93.5 = 0.0359$ ft³
2. $\gamma_t = 4.66/0.0359 = 129.7$ pcf
3. $\gamma_d = 129.7/1.118 = 116.0$ pcf
4. $RC = 116.0/121.0 = 95.9\%$ (≥ 95% required); w = 11.8% is +1.3% of opt (within −1/+3) → passes

Final answer

95.9%, passes

Why the other values are wrong

- **107.2%, passes** — used moist unit weight (forgot ÷(1+w))
- **94.5%, fails** — multiplied by (1 − w) instead of dividing by (1 + w)
- **104.3%, passes** — inverted the ratio

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.9.1–3.9.2 Laboratory and Field Compaction, pp. 146–148; §3.8.3 Weight-Volume Relationships

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v9reh5FJnUc, https://www.youtube.com/watch?v=v_1tQfxBowQ (no creator values copied).

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